

UNIT - 3

SPM

Syllabus :-

Activity planning and Risk Management :-

Objectives of activity planning, project schedules, activities, sequencing and scheduling, network planning models, Formulating network model, forward Pass & Backward Pass techniques, critical path (CPM) method, Risk identification, assessment, Risk planning, Risk management, PERT technique, monte Carlo simulation, resource allocation, creating of critical paths, Cost schedules.

Activity Planning :-

① Activity planning is the process of deciding what activities need to be done, when they should be done, how they will be done, and who will do them in order to achieve a specific goal.

② In simple words,

It is a step-by-step plan of actions to complete a task or project successfully.

* Objectives of activity planning :-

The objectives of activity planning are :-

1.) To achieve goals properly :-

Activity planning helps in completing tasks in a systematic way so that goals are achieved successfully.

2.) To use time effectively :-

It helps in managing time wisely and completing work on time without delay.

3.) To organize work :-

It divides work into small steps, making it easier to understand and perform.

4.) Avoid confusion :-

Proper planning gives clear direction, so there is less confusion about what to do next.

5.) Improve efficiency :-

When activities are planned, work becomes faster and more effective.

6.) Make better decisions :-

Planning helps in choosing the best way to complete a task.

7.) Track progress :-

Helps in checking whether the work is going as planned or not.

Project Schedule :-

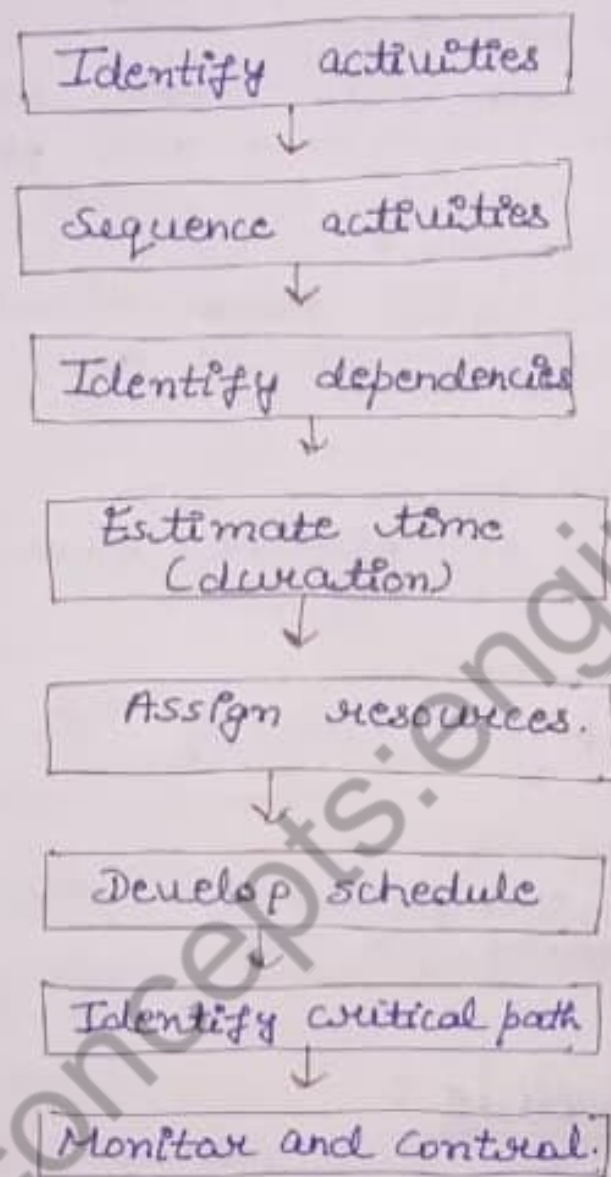
⊙ A project schedule is a time-based plan to complete a project step by step.

⊙ A project schedule is a detailed plan that defines when and in what sequence project activities will be performed.

⊙ It ensures the project is completed on time and in an organized manner.

* Steps of Project schedule :-

The steps to create a project schedule are as :-



1.) Identify activities :-

- Break the project into small tasks (activities).
- This is usually done using work Break-down structure (WBS).
- Each activity should be clear and manageable.

2.) Sequence activities :-

- Arrange all activities in the correct order.

- Decide which task comes first and which comes later.

3.) Identify dependencies:-

- Determine the relationships between tasks.
Ex:- Coding starts only after Design is completed.

4.) Estimate Time:-

- Calculate how much time each activity will take.
- Method:-
Expert Judgement, Past Experience.

5.) Assign Resources:-

- Allocate the resources like:
Human (developers, testers)
Tools (software, hardware).
Budget.

6.) Develop Schedule:-

- Create the actual schedule.

7.) Identify critical path:-

- The longest path in the project network.
- It determines the minimum project completion time.
- Any delay in critical path = delay in entire project.

8.) Monitor and control:-

- Track project progress regularly.
- Compare actual vs planned work.
- Make adjustments if delays occur.

Sequencing and scheduling :-

* Sequencing :-

① Sequencing means arranging project activities in the correct order.

In simple words,

It tells which task should be done first, second, third etc.

② It identifies dependencies between tasks.

③ Helps answer : "Which task comes before or after another?"

* Types of Dependencies :-

1.) Finish-to-start (FS) :- Task B starts after Task A finishes (most common).

2.) Start-to-start (SS) :- Task B starts when Task A starts.

3.) Finish-to-finish (FF) :- Task B finishes when Task A finishes.

4.) Start-to-finish (SF) :- Rare; Task B finishes when Task A starts.

Ex :-

Requirement Analysis → Design → Coding → Testing

Here, each step depends on the previous one.

* Scheduling :

Scheduling is the process of assigning timeframes (start and finish dates) to each activity in the project.

- ① Determines when each task will occur.
- ② Scheduling means deciding when each task of a project will start and finish.

Ex :-

Suppose you are making a software project

- Requirement analysis → 2 days.
- Design → 3 days.
- Coding → 5 days.
- Testing → 3 days.

Now, you fix dates :

- Analysis : → 1-2 June
- Design : → 3-5 June
- Coding : → 6-10 June
- Testing : → 11-13 June.

This complete time plan is called scheduling.

- ① It decides start time and end time of each activity.
- ② It helps to complete project on time.
- ③ It uses time, resources and task order.

Network Planning models :-

Network planning models are techniques used to plan, schedule, and control project activities using a diagram (network).

In simple words :-

- ① It is a graph / diagram that shows all project tasks and their order (sequence).
- ② It shows how all tasks are connected and in what order they will happen.

* Types of Network models :-

There are mainly two types of network models :-

- 1.) Activity - on - Arrow (AOA).
- 2.) Activity - on - Node (AON).

1.) AOA :-

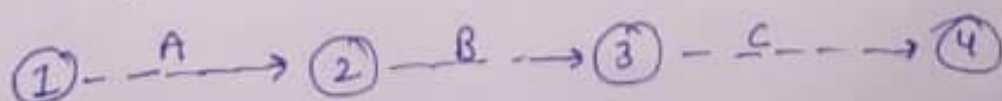
In AOA, activities are represented by arrows and nodes represent events (start/end points).

Arrows $\rightarrow$ activities.

Nodes (circles) $\rightarrow$ Events (start/end points).

Ex :- Let's say :-

- Activity A : Requirements gathering
- Activity B : Design (depends on A)
- Activity C : coding (depends on B)
- In AOA, it looks like this :-



Each arrow (A, B, C) is an activity.
Each number is a node or event.

2.) Activity-on-node (AON)

In this method, activities are represented by nodes (boxes), and arrows show dependencies between tasks.

- Nodes (boxes) = Activities.
- Arrows = Logical relationships or dependencies.

Ex :-

Using the same tasks:



Each box is an activity, and the arrows show the flow of work.

Diffⁿ b/w AOA and AON :-

Feature	AOA	AON
Activities shown as	Arrows	Nodes (Boxes).
Dummy activities	Nodes (circles required sometimes)	Not needed.
Simplicity	more complex	Easier to understand
Events shown as	Nodes (circles)	Not explicitly shown.

Forward Pass and Backward Pass :-

These are techniques used in network diagrams (PERT/CPM) to calculate project timing.

1.) Forward Pass :-

Forward Pass is used to calculate the Earliest Start Time (EST) and Earliest Finish Time (EFT) of activities.

* Steps :-

- Start from the beginning (left to right).
- Take start time = 0.
- Calculate EFT for each activity.
- Move forward through the network.

$$\boxed{EFT = EST + \text{duration}}$$

2.) Backward Pass :-

Backward pass is used to calculate the Latest Start Time (LST) and Latest Finish Time (LFT).

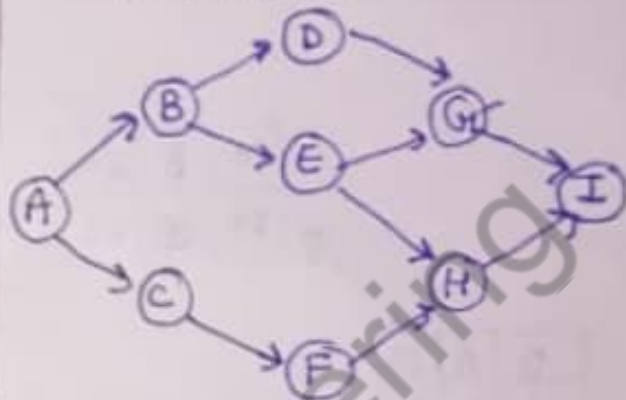
* Steps :-

- Start from the end (right to left).
- Take end time = total project time.
- Calculate LST for each activity.
- Move backward through the network.

$$\boxed{LST = LFT - \text{duration}}$$

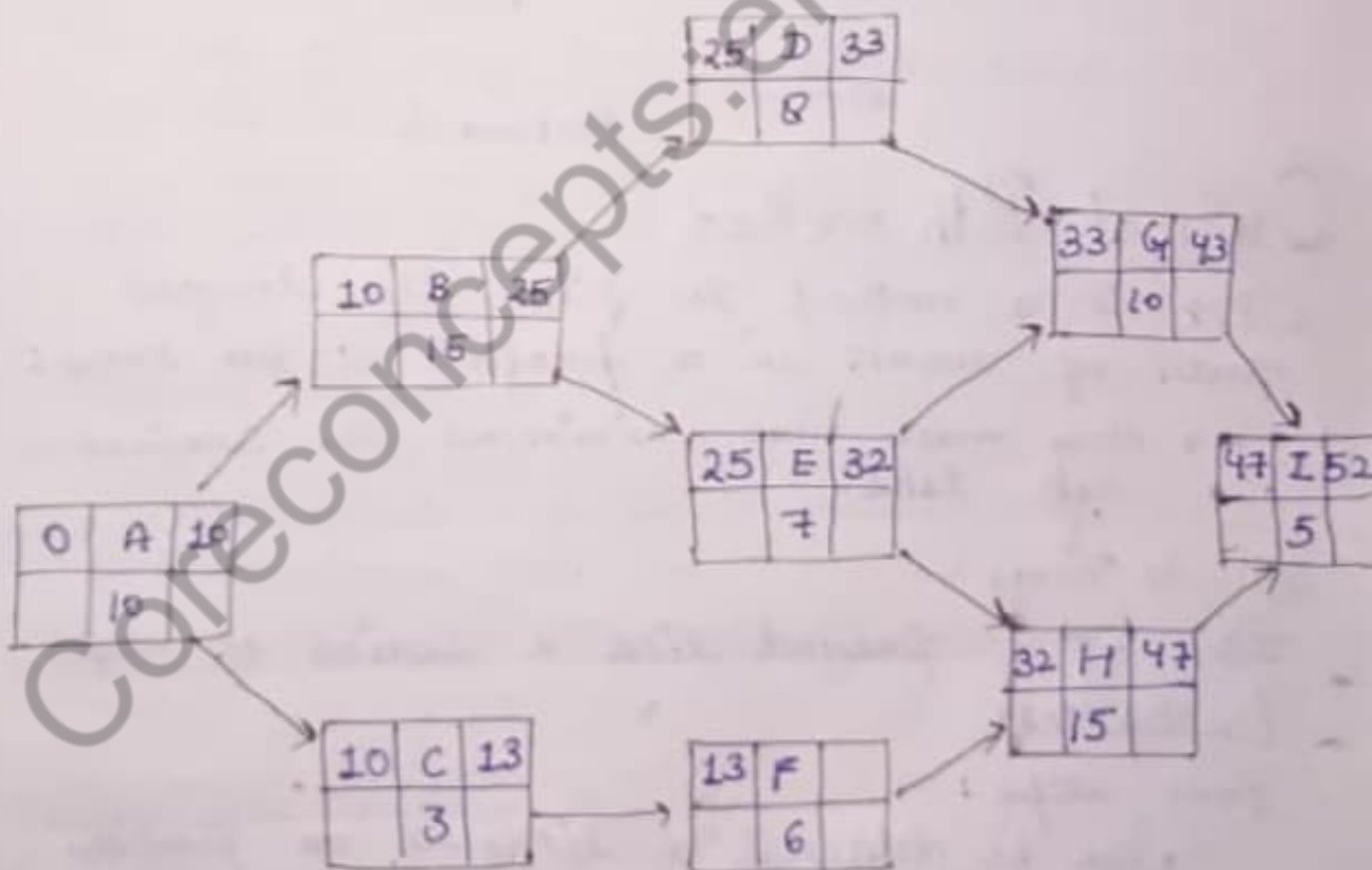
Activity	Predecessor	Duration (Days)
A	-	10
B	A	15
C	A	3
D	B	8
E	B	7
F	C	6
G	E, D	10
H	E, F	15
I	H, G	5

Arrow diagramming method :-

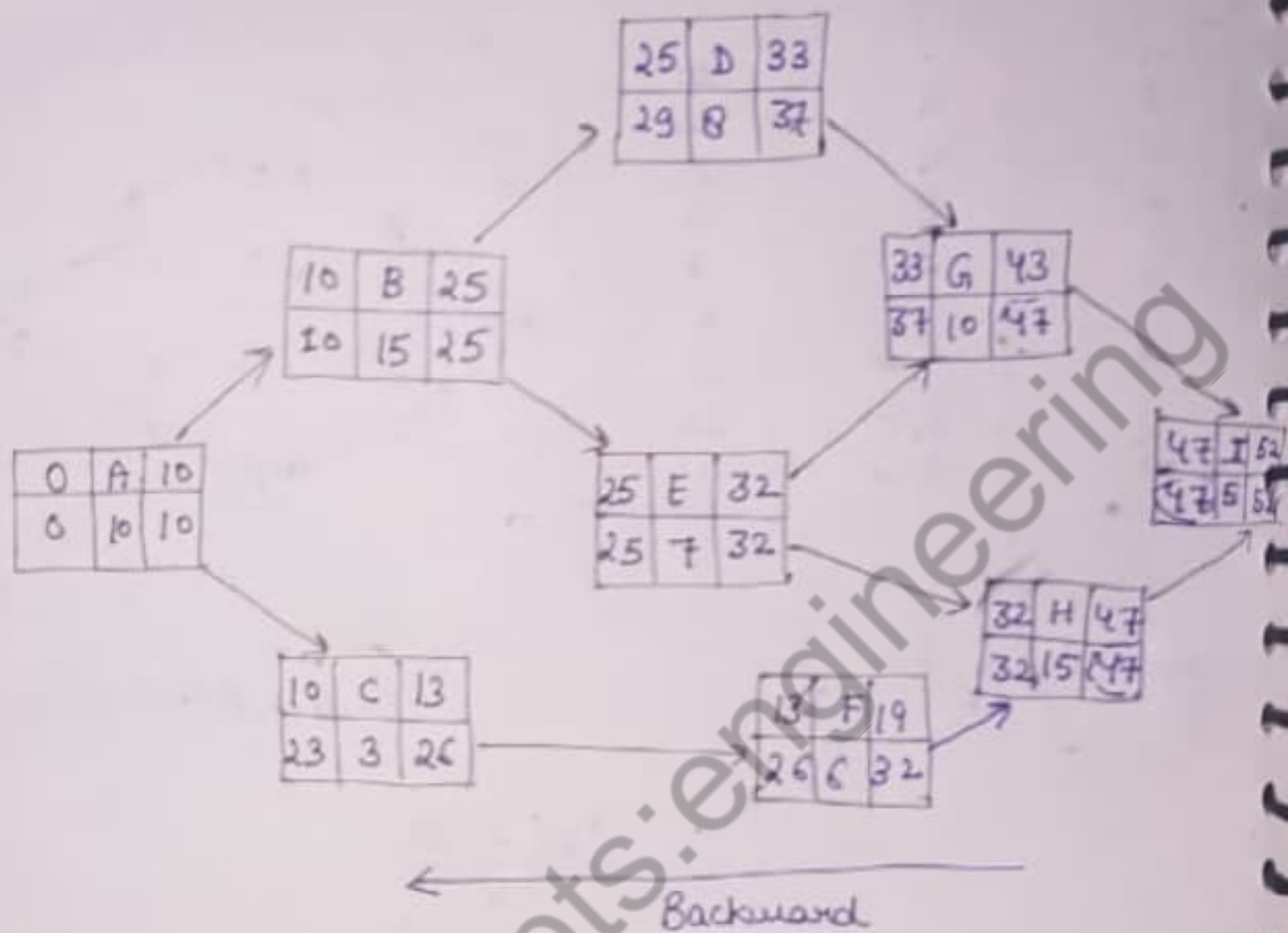


Forward Pass :-

Early start	Activity	Early finish
Late start	Duration	Late finish



Backward Pass :-



Critical Path method :-

CPM is a method to find the longest path of work in a project so we know:

→ How much time (minimum) the project will take.

flexible means :-

Think of a project like a series of steps (activities)

Some steps :

- Can be delayed a little → no problem.
- But some steps :
cannot be delayed at all.

These important steps from the critical path (Numerical on critical path is on the last page).

Cost schedules :-

A cost schedule is a plan that shows how much money will be spent on a project and when it will be spent during the project timeline.

It means :-

Instead of only knowing total cost, we also decide :

- How much money is needed.
- At which stage of the project.

Ex :-

Project duration = 3 months

- Month 1 $\rightarrow$ ₹20,000
- Month 2 $\rightarrow$ ₹30,000
- Month 3 $\rightarrow$ ₹25,000

This distribution of cost over time = Cost schedule.

$\rightarrow$ Combines cost + time.

$\rightarrow$ Helps in budget planning.

Risk Identification :-

Risk Identification means finding all possible problems that can affect a software project before they actually happen.

It answers:

"What things can go wrong in our software project"?

Risk Identification = "find problems before they happen".

* Key features of risk identification :-

1.) Systematic process :-

Risks are not guessed randomly - they are identified using structured methods.

2.) Covers all types of risks :-

- Financial risk.
- Operational risk.
- Strategic risk.
- External risk.

3.) Includes both threats and opportunities :-

While most risks are negative, some can be positive (opportunities), like unexpected market demand.

* Types of Risk :-

In a software project, risks are usually grouped so we can understand and manage them better.

The main types are :-

1.) Technical Risks :-

Related to technology, design, and implementation.

- ⊙ Directly affects software quality and performance.

2.) Project risk :-

Risk related to planning, scheduling, resources, and budget of the project.

- ⊙ Affects overall project completion.

3.) People risk :-

Risk related to team members and management involved in the project.

- ⊙ Poor communication can increase problems.

4.) Requirement risk :-

Risk related to client requirements and their understanding.

- ⊙ Misunderstanding leads to rework.
- ⊙ Affects project scope and correctness.

5.) Operational risk :-

Risk related to day-to-day system operation and maintenance.

- ⊙ Happens after or during system use.

Risk Assessment :-

Risk assessment is the process of analyzing identified risks.

- ⊙ Risk assessment means checking how serious a problem (risk) is before it happens.

Simple Idea :-

After finding risks, we ask :

- "Will this happen?" (Probability)
- "If it happens, how bad will it be?" (Impact).

$$\text{Risk} = \text{Probability} \times \text{Impact}$$

* Key components of risk assessment :-

1.) Probability :-

How likely the risk is to happen.

- High (very likely)
- Medium
- Low (low).

2.) Impact :-

How much effect it will have if it happens

- High (major loss/delay).
- Medium
- Low (minor issue).

This means :-

- A risk with high probability and high impact = very dangerous.
- A risk with low probability and low impact = less important.

* Steps (process) :-

1. Take risks from risk identification.
2. Evaluate their probability.
3. Evaluate their impact.
4. Rank / prioritize risks.
5. focus on the most critical risks.

Risk Planning :-

Risk planning means deciding in advance what you will do if a problem (risk) happens.

Simple Idea :-

After:

- finding risks
- checking how serious they are

Now we ask:

"If this problem comes, what is our plan?"

Ex :-

Suppose you are making a project.

- Risk: Laptop may crash.
- Risk planning:
 - Save work in cloud (Google drive)
 - Keep backup on pen drive.

So if laptop crashes, your work is safe.

* 4 simple ways to handle risk :-

1.) Avoid :- Don't take the risk.

(Ex :- Don't use risky software).

2.) Reduce :- Make the risk smaller.

(Ex :- Do regular testing).

3.) Transfer :- Give risk to someone else.

(Ex :- Take insurance).

4.) Accept :- Be ready if it happens.

(Ex :- Small delay is okay).

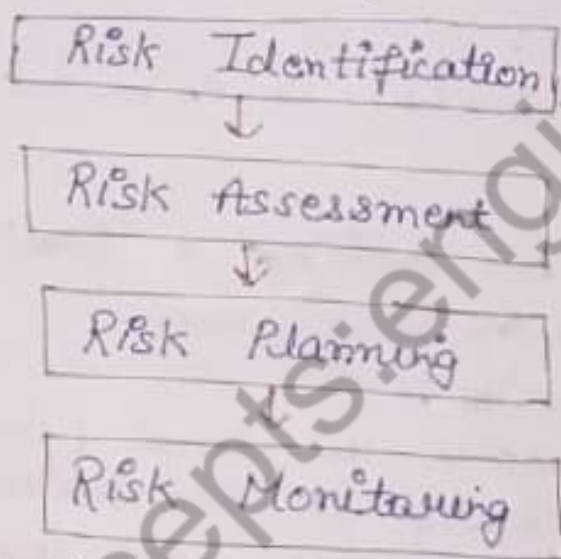
Risk management :-

Risk management means handling problems (risks) in a planned way so they don't harm your project or work.

It means :-

"Find problems early and handle them properly so the project does not fail."

* Steps of risk management :-



1.) Risk Identification :-

- find all possible risks.
- ex :- delay, bugs, unclear requirements.

2.) Risk Assessment :-

- check how serious each risk is
- Based on :

Probability (chance).

Impact (damage)

3.) Risk planning :-

- Decide what action to take.
- Ex : testing, backup, extra resources.

4.) Risk monitoring:

- continuously track risks during project
- check if new risks appear.

Ex :-

Project : software development

- Risk : Team member leaves.
- Assessment : Medium chance, high impact.
- Planning : Keep backup developer.
- Monitoring : Track team performance.

This complete process = Risk management.

PERT Technique :-

- ① PERT (Program Evaluation and Review Technique) is a method used to plan, schedule, and control a project.
- ② A method to estimate project time when exact time is not known.
- ③ PERT helps answer :
"How long will the project take if time is not fixed?"

* Key concept :-

- In PERT, we don't just one time - we take three estimates :
 - Optimistic Time (T_o) :-
Minimum time (best case).
 - Most likely Time (T_m) :-
Normal Time.
 - Pessimistic Time (T_p) :-
Maximum time (worst case).

$$T_e = \frac{T_o + 4 T_m + T_p}{6}$$

This gives the expected time (T_e) of an activity.

Ex :-

Suppose :

$T_o = 2$ days

$T_m = 4$ days

$T_p = 8$ days

Expected time :-

$$= \frac{(2 + 4 \times 4 + 8)}{6}$$

$$= \frac{26}{6}$$

$$\approx 4.33 \text{ days}$$

PERT = "Three time estimate technique for uncertain projects".

Monte Carlo Simulation :-

Monte Carlo simulation is a technique used to analyze risk and uncertainty by running a model many times using random values to find possible outcomes.

Simple Idea :-

Instead of using one fixed value, we ask :

"What are all the possible results if inputs keep changing?"

So, we :

- Take a range of values.
- Generate random values.
- Run the process many times.
- Study the results.

Numerical Example :-

A project activity takes :

- Minimum time = 10 days
- Maximum time = 20 days

We will simulate 5 runs using random values :

Run	Random values (0-1)	Time (days)
1	0.2	12
2	0.5	15
3	0.8	18
4	0.3	13
5	0.7	17

$$\begin{aligned}\text{Average} &= \frac{(12+15+18+13+17)}{5} \\ &= \frac{75}{5} \\ &= 15 \text{ days.}\end{aligned}$$

Most expected time = 15 days.

Possible range = 12 to 18.

It uses many random values and simulation.

Resource allocation:-

Resource allocation is the process of assigning available resources (like people, time, money, and tools) to diff. tasks in a project.

It means:-

Who will do what work, using which resources, and at what time?"

* Types of Resources:-

1.) Human resources:- developers, testers and managers.

People who work on the project.

2.) Time resources:- The amount of time required to complete project tasks.
Each task has a fixed schedule.

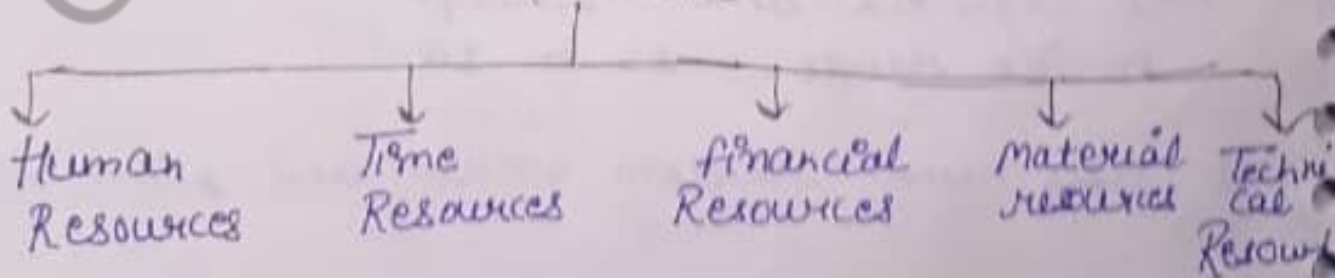
3.) Financial resources:- Money required to complete the project.
Used for salaries, tools and infrastructure.

4.) Material resources:- Physical items needed in the project.

5.) Technical resources:- Technology and tools used in the project.

Includes software tools, programming tools.

Resources



Creating of critical paths :-

Creating a critical path means finding the longest path in a project network diagram.

It determines the minimum time required to complete the project.

It means :-

"The longest path of activities that decides total project time."

In simple words :-

"Which sequence of tasks will take the most time?"

That path = critical path.

① It means :

"Make a project diagram and find the longest route of activities."

* Steps :-

1.) List all activities :-

Write all tasks involved in the project.
(Ex :- A, B, C, D, ...)

2.) Decide sequence :-

find which task depends on which.
What comes first, what comes next.

3.) Draw network diagram :-

Show activities with arrows (flow from start to end).

4.) Assign time :-

Give duration to each activity.

Ex :- A = 3 days

B = 5 days

5.) find all paths :-

Trace all possible paths from start to end.

6.) Calculate total time of each path :-

Add durations of activities in each path.

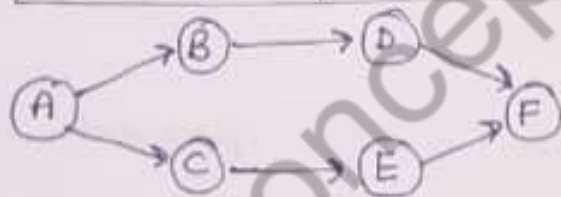
7.) Identify longest path :-

Path with maximum total time = Critical Path.

Numerical :-

Do CPM (Critical path method) analysis and identify the critical path of project.

Activity	Predecessor	Duration
A	-	5
B	A	4
C	A	5
D	B	6
E	C	3
F	D, E	4



Network diagram.

ES	D	EF
	A	
LS	SL	LF

ES → Early start

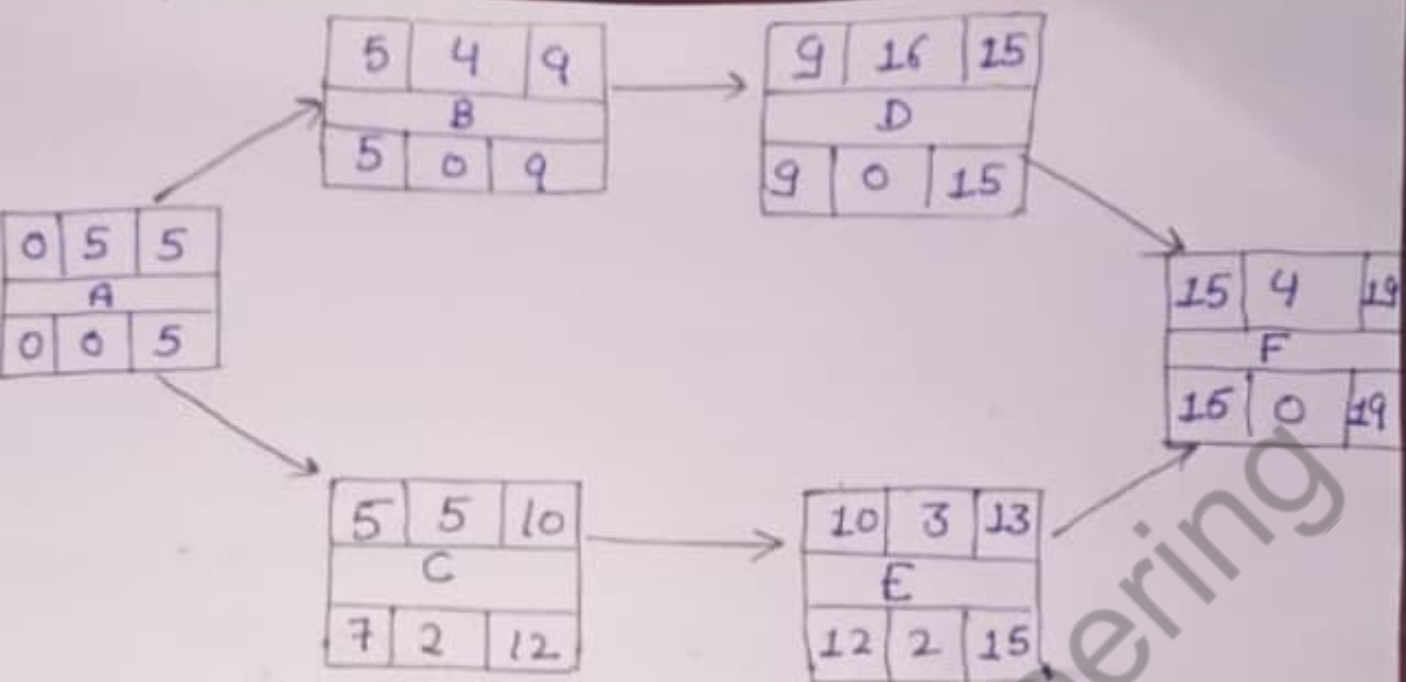
EL → Early

LS → Late start

LF → Late finish

D → Duration

SL → ~~Slack~~
Slack



A B D F

A → B → D → F (Critical path).

— * — * — * — ...

SUBSCRIBE

Careconcepts : engineering